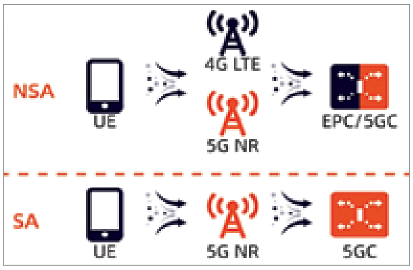
5G in manufacturing: Bosch Automotive Electronics, India's first private 5G network (Source: https://www.privatelteand5g.com)



5G NSA and SA architectures (Source: https://www.iplook.com)

For example, virtual and augmented reality (VR and AR) will enable sports enthusiasts to get a real-time immersive experience; the e-games aficionados will be teleported to an immensely thrilling environment to experience the game in real-life 3D situation, and holographic telepresence will enable teachers' images beamed to remote area classrooms, making them visible simultaneously at multiple locations.

5G in India: On a roll today

The Prime Minister launched the 5G services on 1st October 2022. Airtel and Jio soon commenced rolling out their 5G services in some cities with pan-India rollout slated for December 2023.

Airtel has chosen non-standalone (NSA) 5G networks using the existing 4G networks, while Jio is moving straight to standalone (SA) 5G networks. The key difference between NSA and SA networks is that while an NSA network connects to the existing 4G network to deploy 5G services, the SA architecture needs to implement a 5G core network.

The NSA is faster, easier, and costs less than setting up the SA from the scratch. However, SA deployment makes possible a host of modern and advanced use cases, ensures lower latency—a key next-gen capability, and enables rapid customisation of services and business strategies to maximise return on investments (RoI).

Currently, 72 countries have rolled out 5G services. Most of these are facilitated via 5G NSA, while 24 countries are using 5G SA networks.

The Indian government has set up eight working groups to draw up plans to improve delivery of schemes and services. A 'white paper' has been circulated by the Telecom Ministry on opportunities for designing, developing, and prototyping 5G applications.

Eleven economic verticals have been identified for taking up designing, developing, and prototyping 5G application and use cases. These verticals include city water resources and health; education; sports; infotainment; skill development; tourism; power and resource management; agriculture, fisheries and mining; logistics; intelligent transport systems; railways; ports, airports and commercial hubs; and industry management. Some of the applications being developed are intelligent transport system; city lighting system; wearable telemetry; immersive learning solutions for students; smart grid management; electric metering services; fraud detection; enhancing safety in transportation; autonomous driving systems; automated inventory systems; and remote management of mines.

Awareness programmes and training modules have been drawn up and rolled out for bureaucrats, and faculty and students of engineering and training institutes. More such programmes and modules are being planned and developed. Companies like TCS, Infosys, Wipro, Microsoft, Capgemini, Sensorise, Arkins Labs, and Passenger Drone Research are working intensely on various 5G enabled 'smart' solutions across sectors.

Cutting edge applications

The technologies are ushering in revolutionary changes by making available better quality video calls, instant movie downloads, online high-speed gaming with delightful special effects, and enabling companies to develop layers of applications in the fields of education, healthcare, agriculture, entertainment, automobiles, transportation and logistics, sports and events, infra-

Metaverse is changing the way events are being held and attended (Source: https://sweap.io)